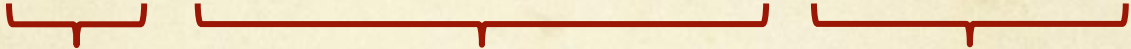


World Wide Web (*WWW*)

- ◆ From user perspective, Web is vast collection of pages
- ◆ Pages named with Uniform Resource Locators (*URLs*)
 - ◆ Example: <http://www.phdcomics.com/comics.php>

The diagram shows three horizontal curly braces under the URL. The first brace is under 'http' and labeled 'Protocol'. The second brace is under 'www.phdcomics.com' and labeled 'Server'. The third brace is under '/comics.php' and labeled 'Page on server'.

Protocol *Server* *Page on server*
- ◆ View of page on browser is result of *conversation* between browser (*client*) and web server (*server*)
 - ◆ *Communication* language: Hypertext Transfer Protocol (*HTTP*)

Hypertext Transfer Protocol (*HTTP*)

- ◆ *Request-response* protocol that normally runs over *TCP*
- ◆ For each request
 - ◆ Establish *connection* between browser & server
 - ◆ *Negotiate* session parameters/settings
 - ◆ Provide for orderly *transfer* of requested content
 - ◆ *Close* connection with server
- ◆ *Stateless* protocol
 - ◆ Does not maintain state across requests

Hypertext Transfer Protocol (*HTTP*)

- ◆ Request message
 - ◆ *Initial request line*: method + local path + HTTP version
 - ◆ *Request header*: information about request
 - ◆ Acceptable content types in response
 - ◆ Acceptable human language in response
 - ◆ Email address of requestor
 - ◆ ...
 - ◆ Blank line
 - ◆ Optional *message body*
 - ◆ Data entered by user
 - ◆ File being uploaded
 - ◆ ...

Hypertext Transfer Protocol (*HTTP*)

- ◆ HTTP has several request *methods*

Method	Description
GET	Read a Web page
HEAD	Read a Web page's header
POST	Append to a Web page
PUT	Store a Web page
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Connect through a proxy
OPTIONS	Query options for a page

Hypertext Transfer Protocol (*HTTP*)

- ◆ Response message
 - ◆ *Initial response line*: HTTP version + status + reason
 - ◆ *Response header*: information about response
 - ◆ Timestamp
 - ◆ Content type
 - ◆ Content length
 - ◆ ...
 - ◆ Blank line
 - ◆ Optional *message body*
 - ◆ File, query output, etc.

Hypertext Transfer Protocol (*HTTP*)

- ◆ Status → response *code* to tell client how request fared

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

Hypertext Transfer Protocol (*HTTP*)

- ◆ Request to retrieve file at URL
`http://www.somehost.com/path/
file.html`

```
GET /path/file.html HTTP/1.0  
From: someuser@jmarshall.com  
User-Agent: HTTPTool/1.0  
[blank line here]
```

- ◆ Sample response from server

```
HTTP/1.0 200 OK  
Date: Fri, 31 Dec 1999 23:59:59 GMT  
Content-Type: text/html  
Content-Length: 1354
```

```
<html>  
<body>  
<h1>Happy New Millennium!</h1>  
(more file contents)  
.  
.  
.  
</body>  
</html>
```